Madagascar only PCA

2024-11-01

Load gt

```
malagasy_merged <-
  gt_load("../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt")
```

This script creates a PCA of only Malagasy individuals, using the Malagasy dataset, comprising of 233 individuals and 3 loci.

```
count_loci(malagasy_merged)
```

```
## [1] 227410
```

```
nrow(malagasy_merged)
```

```
## [1] 233
```

Count individuals from each group

```
pierron_data <- length(which(malagasy_merged$population %in% c("Mikea", "Vezo", "Temoro")))
pierron_data
```

```
## [1] 66
```

```
nrow(malagasy_merged) - pierron_data
```

```
## [1] 167
```

Set imputed

```
malagasy_merged <- gt_impute_simple(malagasy_merged, method = "mean")
gt_set_imputed(malagasy_merged, TRUE)
```

PCA

```
random_pca <- gt_pca_randomSVD(malagasy_merged, total_var = TRUE)
pcs <- as.data.frame(random_pca$u)
variance <- tidy(random_pca)
```

Digeo Individuals

```
malagasy_sample_sheet <-
  as_tibble(read.csv("../data/raw/data_input/Malagasy_pop_gen_sample_sheet_2015_cleaned.csv"))
diego_groups <- malagasy_sample_sheet %>% filter(District == "Diego")
diego <- malagasy_merged %>%
  filter(sub("RR.*", "", id) %in% diego_groups$Sample.ID)
diego$ethnicity <- diego_groups$Ethnic.group[match(sub("RR.*", "", diego$id), diego_groups$Sample.ID)]
unique(diego$ethnicity)
```

```
## [1] "Anjoaty"            "Antakarana"         "Sakalava Tavaratra"
## [4] "Sakalava"
```

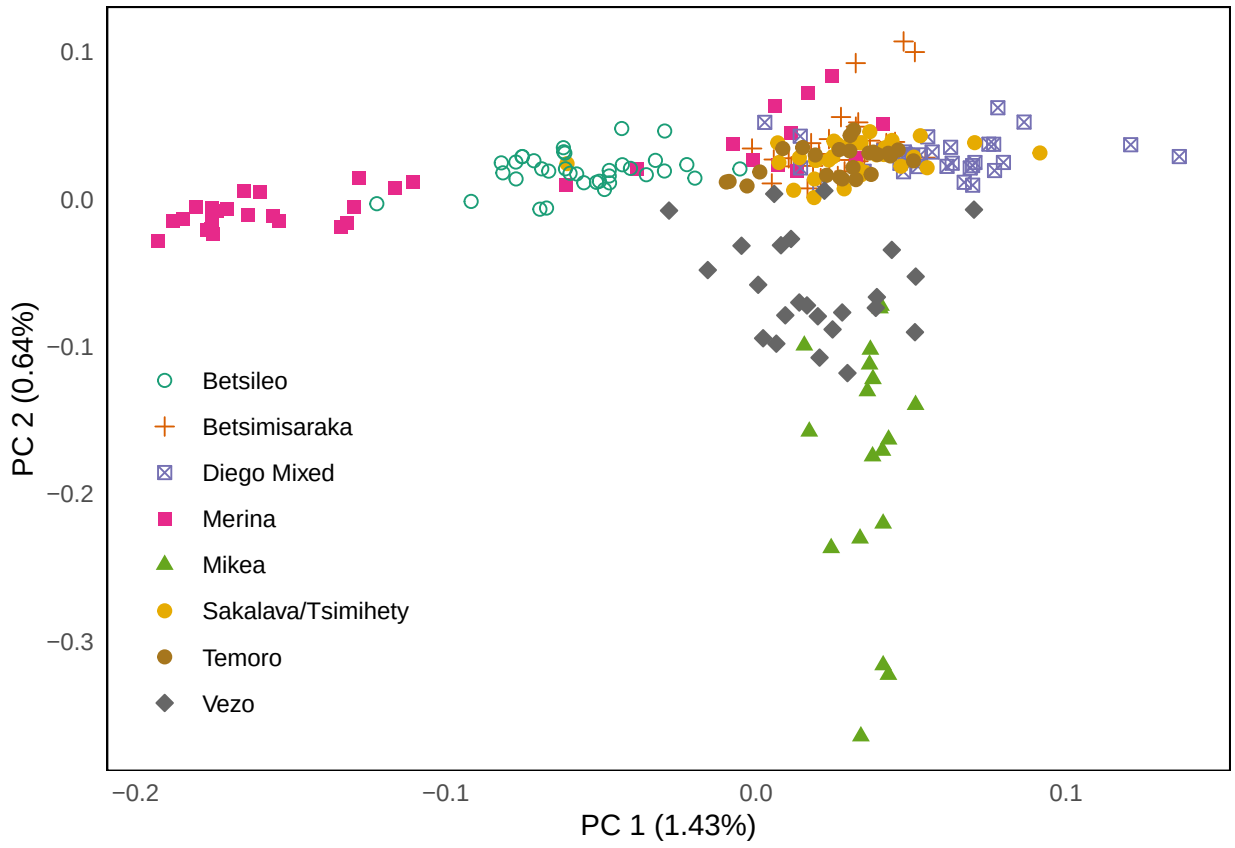
Plot

```
malagasy_merged[malagasy_merged$population == "Sakalava_Tsimihety", "population"] <- "Sakalava/Tsimihety
malagasy_merged[malagasy_merged$population == "Diego", "population"] <- "Diego Mixed"

p1 <- ggplot(pcs, aes(
  x = V1, y = V2,
  group = malagasy_merged$population,
  colour = malagasy_merged$population,
  shape = malagasy_merged$population,
  fill = malagasy_merged$population
)) +
  geom_point(size = 2, alpha = 1) +
  scale_shape_manual(values = c(1, 3, 7, 15, 17, 19, 21, 23)) +
  labs(
    title = NULL,
    x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"),
    y = paste0("PC 2 (", round(variance$percent[2], 2), "%)"),
    colour = NULL, shape = NULL, fill = NULL
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  theme_minimal() +
  theme(
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    legend.position = c(0.15, 0.3)
  )
```

```
## Warning: A numeric ‘legend.position‘ argument in ‘theme()‘ was deprecated in ggplot2
## 3.5.0.
## i Please use the ‘legend.position.inside‘ argument of ‘theme()‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

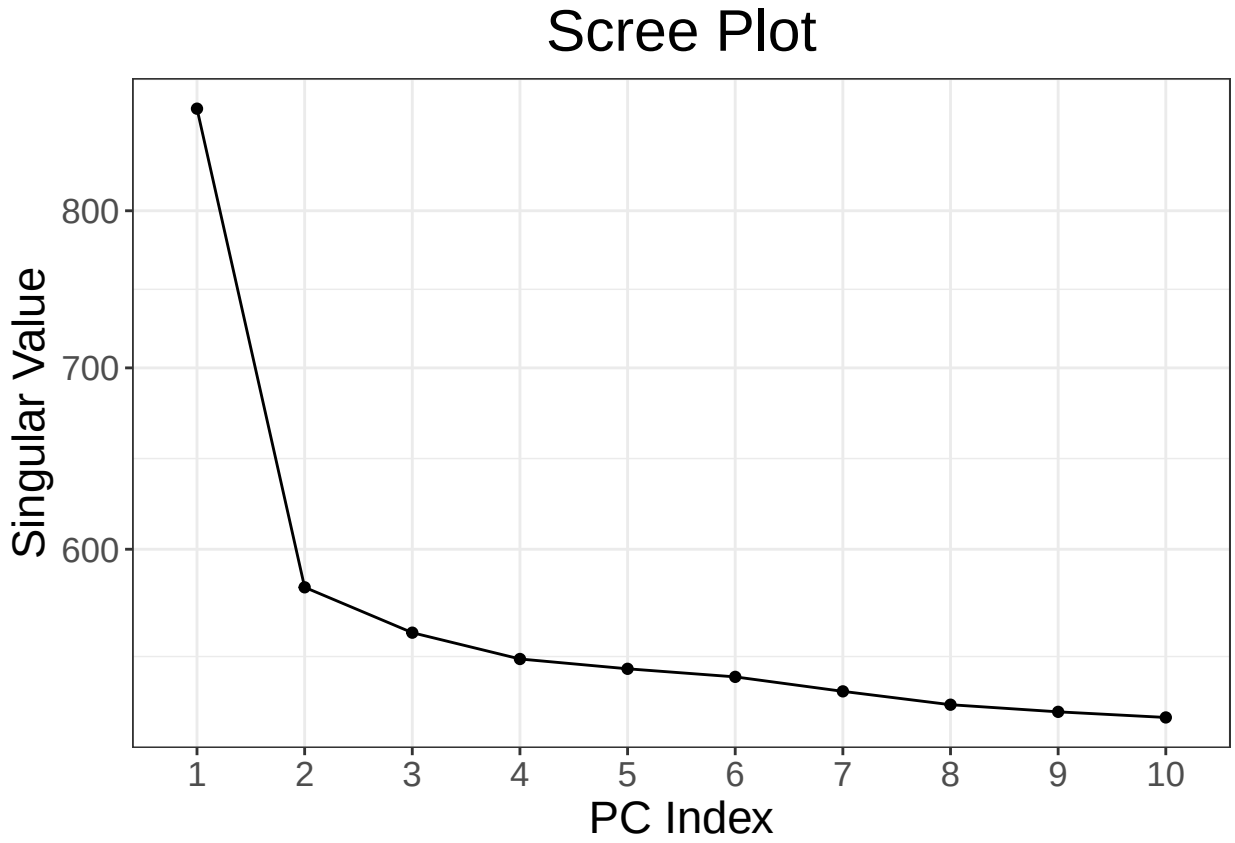
```
p1
```




```
ggsave(paste0(res_dir, "madagascar_pca.jpg"), width = 7, height = 5, dpi = 300)

saveRDS(p1, paste0(res_dir, "madagascar_pca.rds"))
```

```
autoplot(random_pca, type = "scree")
```



This PCA uses the KING relatedness filtering of 0.088, equivalent to second degree relatives Manichaikul et al. (2010). There is justification for keeping second degree close relatives (aunts, uncles, grandparents, grandchildren, nieces, nephews, and half-siblings) because the prevalence of consanguinity in each population is unknown, but likely to be high in places, and therefore removing all second degree relatives may substantially reduce the sample size.

However, altering the relatedness filtering only reduced the sample size by ~10. The PCA plot shows a similar distribution of the Malagasy samples, but with a slightly different spread and slightly reduced explained variance when compared to relatedness filtering with a kinship coefficient of 0.177 (first degree relatives). It therefore appears that the relatedness filtering has a small effect on the PCA, but the overall structure of the Malagasy samples is maintained.

'the KING option ''–unrelated'' proceeds as follows. Related individuals (defined by existing pedigree or estimated kinship coefficient ,0.088) are first clustered into connected groups (i.e., families). Within each family cluster, individuals are ranked according to the count of unrelated family members (having estimated kinship coefficient ,0.022). To construct a set of unrelated individuals, we first select the individual with the largest count of unrelated individuals within the family cluster. We then proceed to choose the individual with the next highest rank (number of unrelated family members) within the family cluster, only if that individual is not related to any of the previously selected individuals in the list of unrelated individuals' Manichaikul et al., 2010)

'–related –degree 2 specifies that only related pairs (up to the 2nd-degree in this case) between families are included in the output. Specifically all pairs across families with a kinship coefficient less than 0.0884 will be excluded from the output.'

Manichaikul, Ani, Josyf C. Mychaleckyj, Stephen S. Rich, Kathy Daly, Michèle Sale, and Wei-Min Chen. 2010. "Robust Relationship Inference in Genome-Wide Association Studies." *Bioinformatics* 26 (22): 2867–73. https://doi.org/10.1093/bioinformatics/btq559.